

Java OOP Tutorial Worksheet

Question Paper

Q1 (2 marks)

Study:

```
SavingsAccount s = new SavingsAccount("Alice",1000);
```

- (a) Identify the class being instantiated.
- (b) How many objects are created?

Q2 (3 marks)

Account is declared as:

```
public abstract class Account
```

- (a) What is an abstract class?
- (b) Can 'Account a = new Account();' compile? Explain.

Q3 (6 marks)

Study the constructors:

```
public SavingsAccount(){ super(); }
```

```
public SavingsAccount(String s,double d){ super(s,d); }
```

- (a) Purpose of super()?
- (b) Purpose of super(s,d)?
- (c) Which constructor finally initializes Nick and 5000?

Q4 (5 marks)

Trace:

```
SavingsAccount a = new SavingsAccount();
```

Given:

```
Account(){ this("Nick",5000); }
```

```
Account(String s,double d){...}
```

- (a) First constructor?
- (b) Next?
- (c) Final values?

Q5 (4 marks)

Why are accountName and balance declared private? How can balance be accessed?

Q6 (4 marks)

withdraw() and withdraw(double amount).

- (a) OOP concept?
- (b) Explain.

Q7 (5 marks)

private static double interestRate=1.5;

(a) Meaning of static?

(b) Which is correct?

A account.setInterestRate(2.5)

B Account.setInterestRate(2.5)

Q8 (6 marks)

SavingsAccount overrides addInterest().

(a) OOP concept?

(b) Explain.

Q9 (8 marks)

Account a = new SavingsAccount("Alice",1000);

a.addInterest();

(a) OOP concept?

(b) Reference type?

(c) Object type?

(d) Which addInterest() runs and why?

Q10 (6 marks)

Complete:

SavingsAccount s=new SavingsAccount("Alice",1000)

Account a=new SavingsAccount("Bob",2000)

Fill reference type and object type.

Q11 (8 marks)

CurrentAccount extends Account and overrides addInterest(). Explain which addInterest() executes for Account references pointing to SavingsAccount and CurrentAccount.

Challenge (10 marks)

Predict output:

Account a=new SavingsAccount("Alice",1000);

print details; withdraw(100); addInterest(); print details.

Also predict output of:

SavingsAccount b=new SavingsAccount();

print details;